Analyzing a Market Test

Part 5 – New Menu Launch Project

ABSTRACT

Throughout this report, I will demonstrate every taken step to came up with a recommendation to whether the Round Roasters chain should launch a new menu or not.

Maged Baheig

Problem Solving with Advanced Analytics-Udacity/Bertelsmann2021

New Menu Launch – Analyzing a Market Test Report

Complete each section. When you are ready, save your file as a PDF document and submit it here

Step 1: Plan Your Analysis

To perform the correct analysis, you will need to prepare a data set. (500 word limit)

Answer the following questions to help you plan out your analysis:

1. What is the performance metric you'll use to evaluate the results of your test?
2. What is the test period?
3. At what level (day, week, month, etc.) should the data be aggregated?

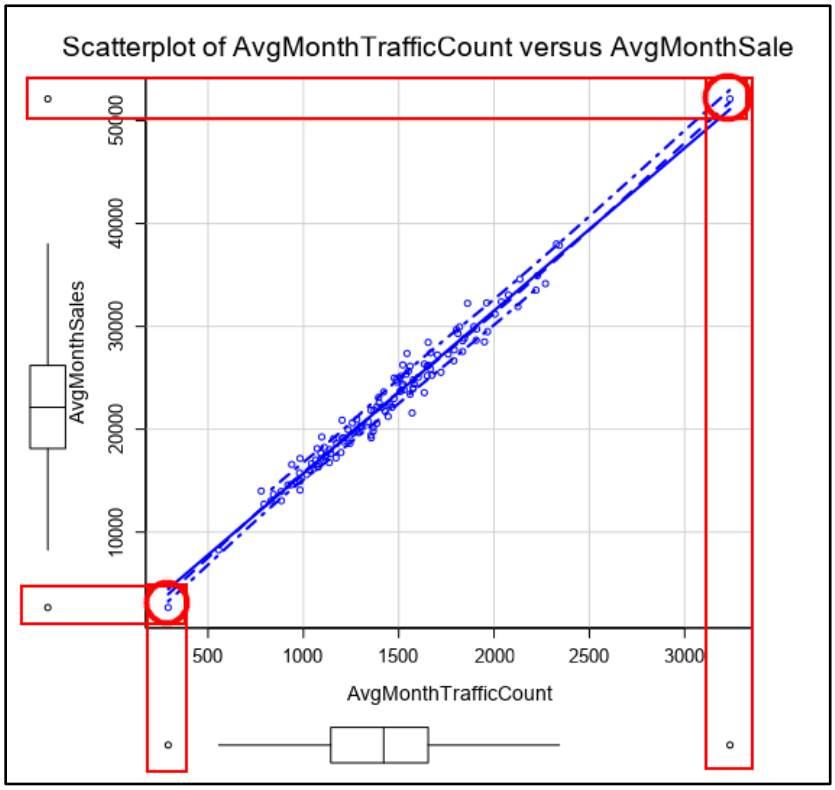
- The **change** which Round Roasters are about to do is "***Offering the updated menu along with television advertising***".
- This change will be made on the **independent variable "The Menu"**.
- We assume that a change in the independent variable "Menu" could have an **impact** on the **dependent variable "Profit".** We need to evaluate that impact.
- **For an impact to be considered as "Practically Significant"**, the predicted impact to profitability should be enough to justify the increased marketing budget: **at least 18% increase** in profit growth compared to the comparative period while compared to the control stores.
- Accordingly,
  1. **Performance metric is**: Sum Gross Margin.
  2. **Test Period**: a total of **12** continuous weeks:
     - *Test Start Date: 2016-April-29*
     - *Test End Date: 2016-July-21*
  3. **Time-based aggregation level is**: Week. Weekly level aggregation.
- Based on above 3, we could name our performance metric as "**Weekly Gross Margin**".

<u>Step 2</u>: Clean Up Your Data

In this step, you should prepare the data for steps 3 and 4.
You should aggregate the transaction data to the appropriate level and filter on the appropriate data ranges.
You can assume that there is no missing, incomplete, duplicate, or dirty data.
You're ready to move on to the next step when you have weekly transaction data for all stores.

- ***<u>Removing Outliers:</u>***
  - *While matching Treatment to their control variables, Alteryx will consider and avoid these outliers*. **<u>However</u>**, **I decided to remove them as a best practice**.
  - I've decided to remove **2 outliers** control stores from the "*round-roaster-stores.csv*" based on:
    1) AvgMonthSale
    2) AvgMonthTrafficCount

#	Store ID	Outlier becasue	AvgMonthSale	AvgMonthTrafficCount
1	**11518**	**High Value**	**52119**	**3238**
2	**9338**	**Low Value**	**2661**	**291**



- ***Data Aggregation Result:***
  - Below is a **snapshot** of the final data shape to be used in "**AB Trend**" Tool.
  - It consists of the **weekly transaction data** for all stores (***76 Week for each store***).

8 of 8 Fields | Cell Viewer | 10,108 records displayed | Search

Record	StoreID	WeekNo	WeekStart	WeekEnd	CountDistinct_Invoice Number	Sum_QTY	Sum_Sales	Sum_Gross Margin
1	10018	1	2015-02-06	2015-02-12	308	1,363	4,741.48	2,212.7105
2	10018	2	2015-02-13	2015-02-19	288	1,365	4,571.25	2,164.007
3	10018	3	2015-02-20	2015-02-26	204	953	3,348.25	1,560.929
4	10018	4	2015-02-27	2015-03-05	320	1,448	5,114.96	2,342.984
5	10018	5	2015-03-06	2015-03-12	284	1,349	4,799.48	2,199.4065
6	10018	6	2015-03-13	2015-03-19	288	1,278	4,554.97	2,103.143
7	10018	7	2015-03-20	2015-03-26	194	876	2,999.55	1,412.927
8	10018	8	2015-03-27	2015-04-02	286	1,292	4,519.61	2,124.3715
9	10018	9	2015-04-03	2015-04-09	274	1,340	4,727.05	2,216.152
10	10018	10	2015-04-10	2015-04-16	215	1,011	3,689.33	1,686.246
11	10018	11	2015-04-17	2015-04-23	277	1,177	4,168.13	1,938.1365
12	10018	12	2015-04-24	2015-04-30	251	1,127	4,002.32	1,874.485
13	10018	13	2015-05-01	2015-05-07	201	945	3,334.79	1,571.1225
14	10018	14	2015-05-08	2015-05-14	207	971	3,417.03	1,586.3945
15	10018	15	2015-05-15	2015-05-21	334	1,505	5,259.41	2,473.113
16	10018	16	2015-05-22	2015-05-28	243	1,116	3,921.49	1,815.0065
17	10018	17	2015-05-29	2015-06-04	321	1,414	5,007.67	2,330.4885
18	10018	18	2015-06-05	2015-06-11	406	1,747	6,148.26	2,897.6665
19	10018	19	2015-06-12	2015-06-18	294	1,345	4,631.32	2,155.198
20	10018	20	2015-06-19	2015-06-25	389	1,658	5,751.79	2,639.836
21	10018	21	2015-06-26	2015-07-02	314	1,385	4,687.75	2,188.6865

<u>Step 3</u>: Match Treatment and Control Units

In this step, you should create the trend and seasonality variables, and use them along with you other control variable(s) to match two control units to each treatment unit. Note: Calculate the number of transactions per store per week to calculate trend and seasonality.

Apart from trend and seasonality...

1. What control variables should be considered? Note: Only consider variables in the RoundRoastersStore file.
2. What is the correlation between your each potential control variable and your performance metric?
3. What control variables will you use to match treatment and control stores?
4. Please fill out the table below with your treatment and control stores pairs:

Treatment Store	Control Store 1	Control Store 2

1. A List of Control Variables Should be Considered

- If a variable could have an impact on the target variable directly or indirectly, we should consider it as a "***potential control variable***" and then check it statistically.
- Considering **data availability constrains**, all below variables should be considered as "***potential control variable***".

  1) AvgMonthTrafficCount.
  2) AvgMonthQTY.
  3) AvgMonthSales.
  4) Sq_Ft.
  5) Region.

8 of 8 Fields | Cell Viewer | 133 records displayed | Target Variable | Search | Data

Record	StoreID	AvgMonthTrafficCount	AvgMonthQTY	AvgMonthGrossMargin	AvgMonthSales	Sq_Ft	Region	Test_Group
1	10018	1,759.5	7,860.75	12,712.250458	27,310.784167	1,183	West	Control
2	10068	1,602.416667	7,150.333333	11,491.490917	25,039.170833	1,198	West	Control
3	10118	1,289.583333	5,816.583333	9,154.246417	19,673.621667	1,204	West	Control
4	10168	1,894.083333	8,688.916667	14,067.897792	29,975.238333	1,195	West	Control
5	10218	1,475.083333	6,657.083333	10,571.701625	22,985.4375	1,193	West	Control
6	10268	2,342.583333	10,737.5	17,694.371708	37,873.661667	1,191	West	Control
7	10318	1,522.083333	7,013.25	11,426.521875	24,367.835	1,147	West	Control
8	10368	1,835.25	8,160.75	13,186.020792	28,596.103333	1,159	West	Control
9	10418	1,803.083333	8,413.666667	13,670.32925	29,683.970833	1,146	West	Control
10	10468	2,075.25	9,436.916667	15,439.846167	33,057.4425	1,151	West	Control
11	10518	1,091.5	5,109.583333	8,146.200333	17,673.12	1,147	West	Control
12	10568	2,220.416667	9,688.416667	15,436.528375	33,528.460833	1,192	West	Control
13	10618	1,559.916667	6,817.666667	10,984.211458	23,396.161667	1,187	West	Control
14	11268	1,109.5	5,227.166667	8,475.062708	18,246.581667	1,328	West	Control

- <u>We should check its correlation statistically with</u>:

  *a) The evaluation metric "**Weekly Gross Margin**".*

  *b) **Itself**.*

- <u>During checking correlation step</u>: I've decided to aggregate the Profit using <u>Average on a monthly level instead of Sum on weekly level</u>, ***just to be at the same Aggregation methods with available control variables***.

- <u>However, during the analysis phase</u>, <u>the profit will be aggregated using Sum on weekly level</u>, ***to get the actual expected lift***.

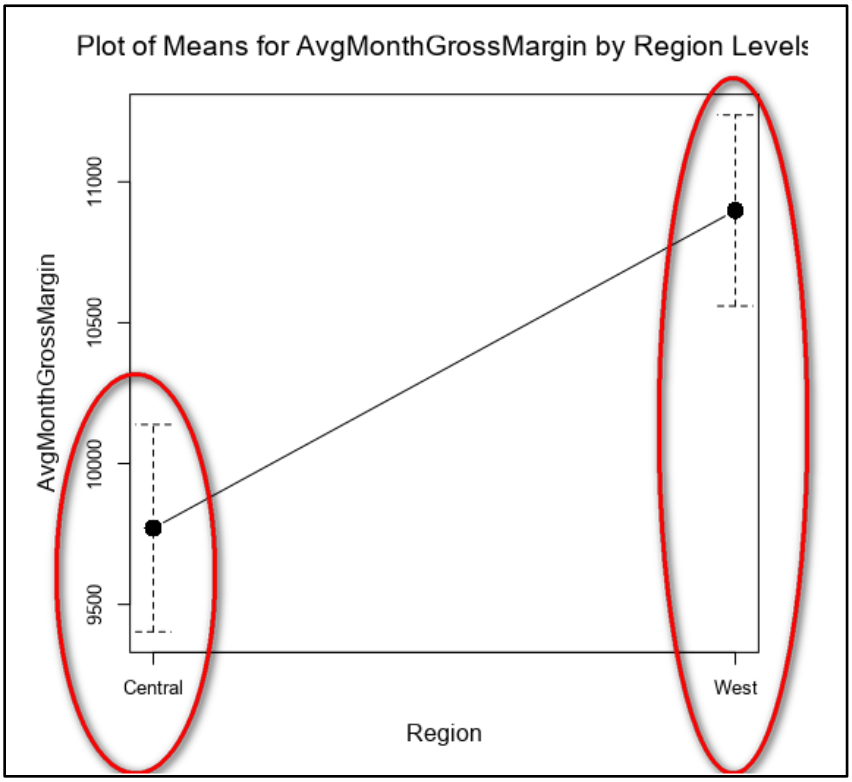
2. Checking Correlation

a) ***Between each potential control variable and the evaluation metric "AvgMonthGrossMargin".***

Pearson Correlation Analysis

Focused Analysis on Field AvgMonthGrossMargin

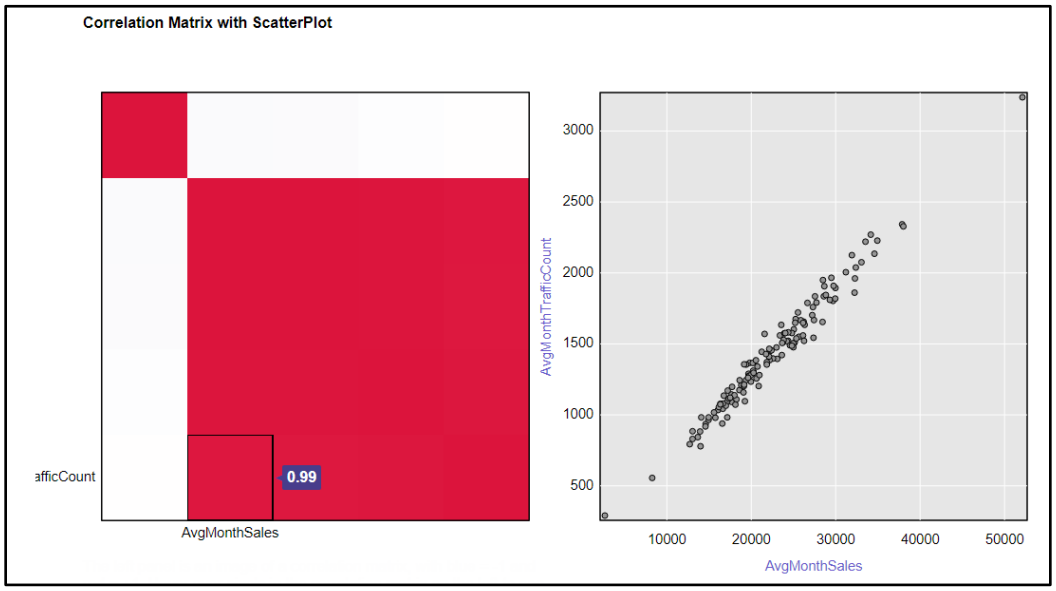
	Association Measure	p-value	
AvgMonthSales	0.999356	0.00000	***
AvgMonthQTY	0.996720	0.00000	***
AvgMonthTrafficCount	0.985095	0.00000	***
Sq_Ft	-0.024255	0.78168	

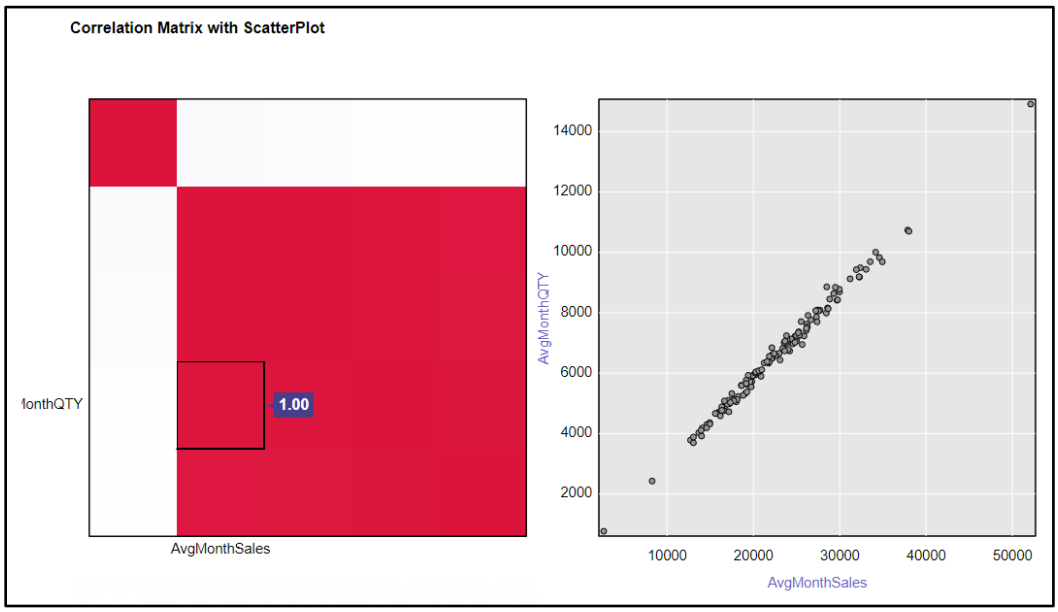


- Below **4 Control variables** have a strong correlation with the Target variable, and should be considered.
  1) **AvgMonthTrafficCount.**
  2) **AvgMonthQTY.**
  3) **AvgMonthSales.**
  4) **Region.**

- **Sq_Ft** variable doesn't have a correlation with the Target variable, so will be ignored.

b) ***Between each potential control variable, the others control variables*** *.*



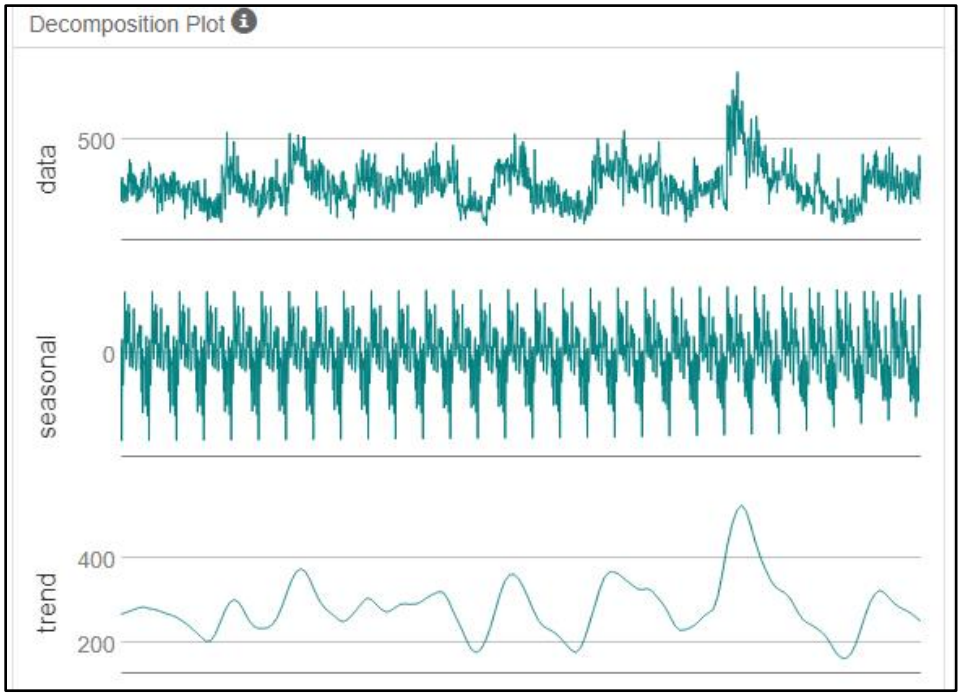


- As clear above, the **3 Control variables** have a strong correlation with each other's.
  1) **AvgMonthTrafficCount.**
  2) **AvgMonthQTY.**
  3) **AvgMonthSales.**

- Accordingly, we should consider **only 1 of them**. The variable which has the **highest correlation association measure value.**
  - **AvgMonthSales** variable [correlation association measure value = **0.999356**].

3. Control variables will be used to match treatment and control stores

- The first control variables we should use are **AvgMonthSales** and **Region**.
- Beside it, I explored the existence of **trend and seasonal effect** using "**TS Plot**" and there are a clear seasonal and trend effect on weekly traffic count, as demonstrated below.



- Accordingly, we should use **Trend** and **Seasonal** effect while matching treatment and control stores.
- I did that using "**AB Trend**" Tool, resulting below the **2 new control variables**:

3 of 3 Fields | Cell Viewer | 133 records displayed

Record	StoreID	Trend	Seasonality
1	10018	0.256985	-0.001456
2	10068	-0.265219	0.009305
3	10118	-0.682138	0.011894
4	10168	-0.450393	0.011854

- **So, we will use the below 4 control variables:**

  1) **Trend.**
  2) **Seasonality.**
  3) **AvgMonthSales.**
  4) **Region**

4. Treatment and Control Stores Pairs

Treatment Store	Control Store 1	Control Store 2
2288	9081	2568
2293	12219	9524
2301	3102	9238
2322	2409	12786
2341	3185	12536
1664	7162	7484
1675	8162	7434
1696	1964	1863
1700	1630	2014
1712	2114	8312

Step 4: Analysis and Writeup

Conduct your A/B analysis and create a short report outlining your results and recommendations. (250 words limit)

Answer these questions. Be sure to include visualizations from your analysis:

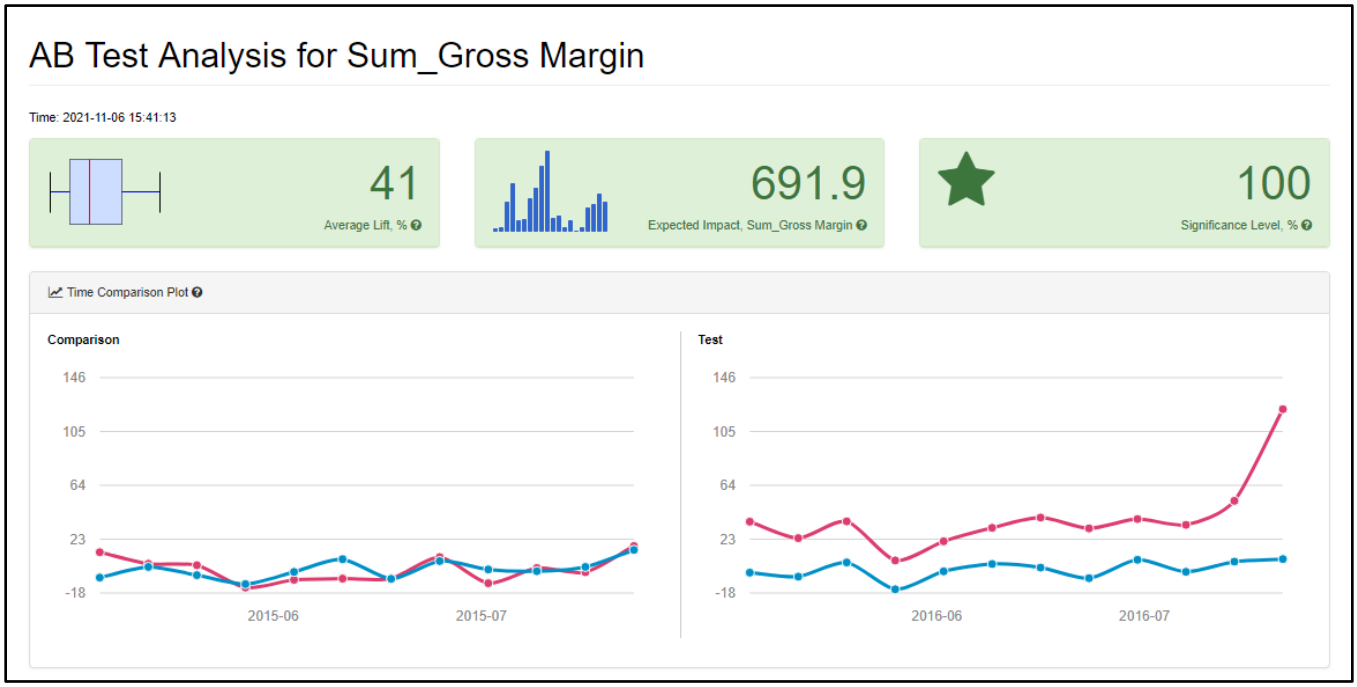
1. What is your recommendation - Should the company roll out the updated menu to all stores?
2. What is the lift from the new menu for West and Central regions (include statistical significance)?
3. What is the lift from the new menu overall?

Before you Submit

Please check your answers against the requirements of the project dictated by the rubric here. Reviewers will use this rubric to grade your project.

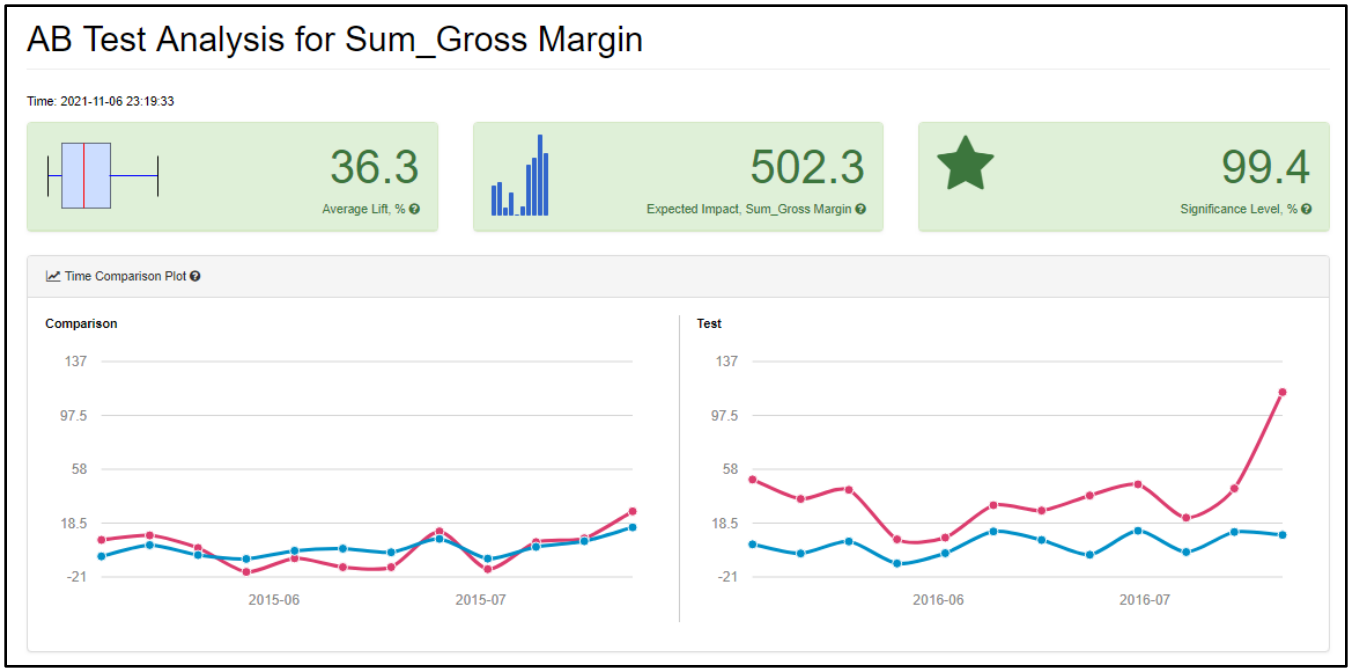
1. **My Recommendation**

- I strongly recommend **launching and applying the new menu changes to all stores**, due to below reasons:
  - **Practically Significant:**
    - The **average gross margin**, per week, in the treatment store, **increased** by **41%**, during the experiment.
  - **Statistically Significant:**
    - **Overall**, we are **100% sure that the result is statistically Significant** [**P-Value = 0.000**].



2. **The Lift from the New Menu**

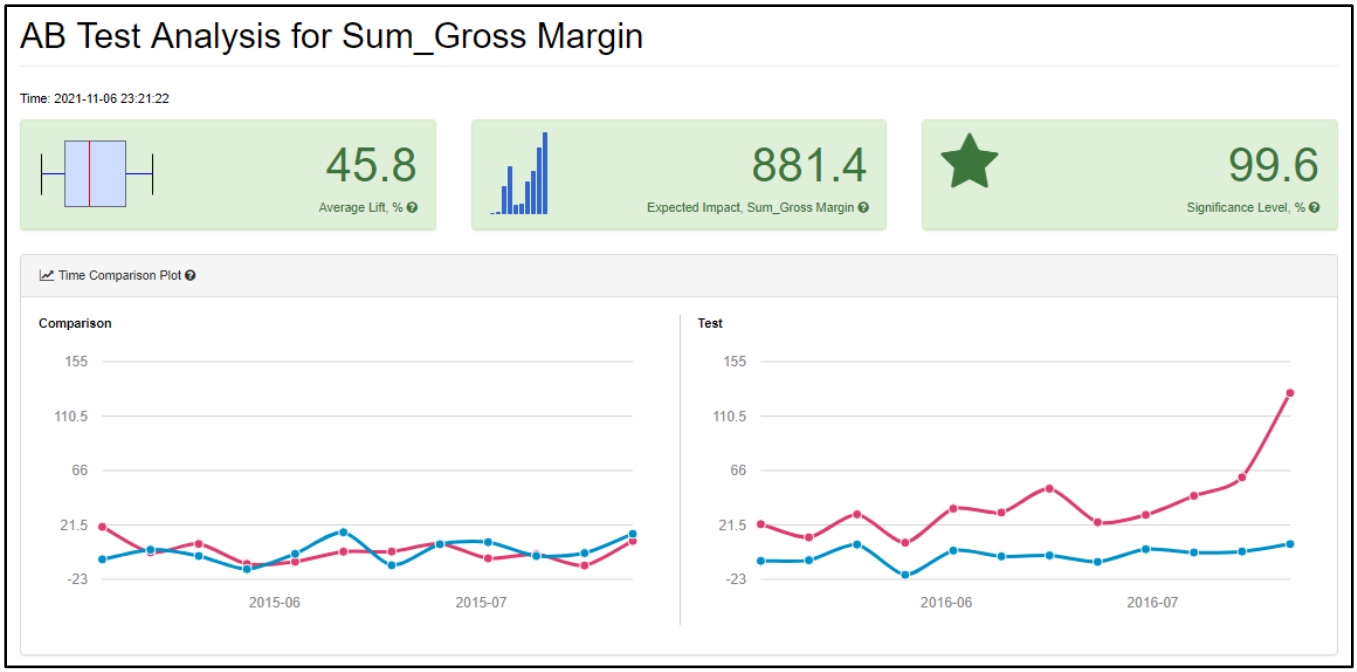
- ***Western Region:***
  - **Practically Significant:**
    - The **average gross margin**, per week, in the treatment store, **increased** by **36.3%**, during the experiment.
  - **Statistically Significant:**
    - We are **99.4% sure that the result is statistically Significant** [**P-Value = 0.006%**].



- **Expected Lift :**
  - **On average**, we expect **502.3$** increase in the **Sum_Gross Margin** per **sore** per **week**.
  - <u>if the treatment were</u> applied, **on average**, we could get an **increased in Western Gross Margin** by **4,879,403$**.

Actual_Sum_Gross Margin	Expected_IncreasedSum_Gross_Margin	Wetern_Gross_Margin_Difference
11,900,982.489	16,780,385.30949	4,879,402.82049

- ***Central Region:***
  - **Practically Significant:**
    - The **average gross margin**, per week, in the treatment store, **increased** by **45.8%**, during the experiment.
  - **Statistically Significant:**
    - We are **99.6%** **sure that the result is statistically Significant**, [**P-Value = 0.004%**].



- **Expected Lift :**
  - **On average**, we expect **881.4$** increase in the **Sum_Gross Margin** per **sore** per **week**.
  - <u>if the treatment were</u> applied, **on average**, we could get an **increased in Central Gross Margin** by **2,019,039$**.

Actual_Sum_Gross Margin	Expected_IncreasedSum_Gross_Margin	Central_Gross_Margin_Difference
4,924,484.484	6,943,523.12244	2,019,038.63844

3. **The Lift from the New Menu Overall**

- <u>if the treatment were</u> applied, **on average**, we could get an **increased in Overall Gross Margin** by **6,898,441$**.

Actual_Sum_Gross Margin	Expected_IncreasedSum_Gross_Margin	Overall_Gross_Margin_Difference
16,825,466.973	23,723,908.43193	6,898,441.45893

Thank You